Exp 4: Data Modelling and Form Design for a Business Application in Zoho Creator

1. Aim

To design and build a functional “Gym Membership Enrollment Form” application in Zoho Creator that demonstrates correct usage of all thirteen Basic Fields and all seventeen Advanced Fields available in the Form Builder, while solving a real fitness-center requirement of collecting, processing, and managing new member enrollments — from first contact through fee computation and gate-access issuance.

2. Procedure

2.1 Problem Statement & Application Scope

A fitness center currently enrolls new members using a printed registration form at the front desk. This results in illegible entries, missing emergency contact details, and no structured record of a member's fitness goals or medical conditions before training begins. Trainers have no visibility into a member's preferences before the first session, and the front desk has no automated way to verify assigned trainers, compute fees after referral discounts, or issue a scannable membership ID for gate access.

The application designed here, Membership_Enrollment_Form, lets a prospective member submit an enrollment request through a single online form. It captures personal identity, contact details, emergency contact, membership plan preference, fitness goals, and a health declaration/waiver using the thirteen Basic Fields, then extends this with the seventeen Advanced Fields to support family enrollment, fee computation with referral discounts, trainer-assignment verification against a master Trainers table, and automated membership-ID / QR-code generation for gate access.

2.2 Data Model & ER Diagram

The diagram below shows the internal schema built for this application: the primary form entity with its Basic and Advanced fields, the Family_Members_Enrolled SubForm capturing repeating family rows, the Trainers_Master table the Assigned Trainer Lookup field draws from, and the choice sets backing the form's picklists.

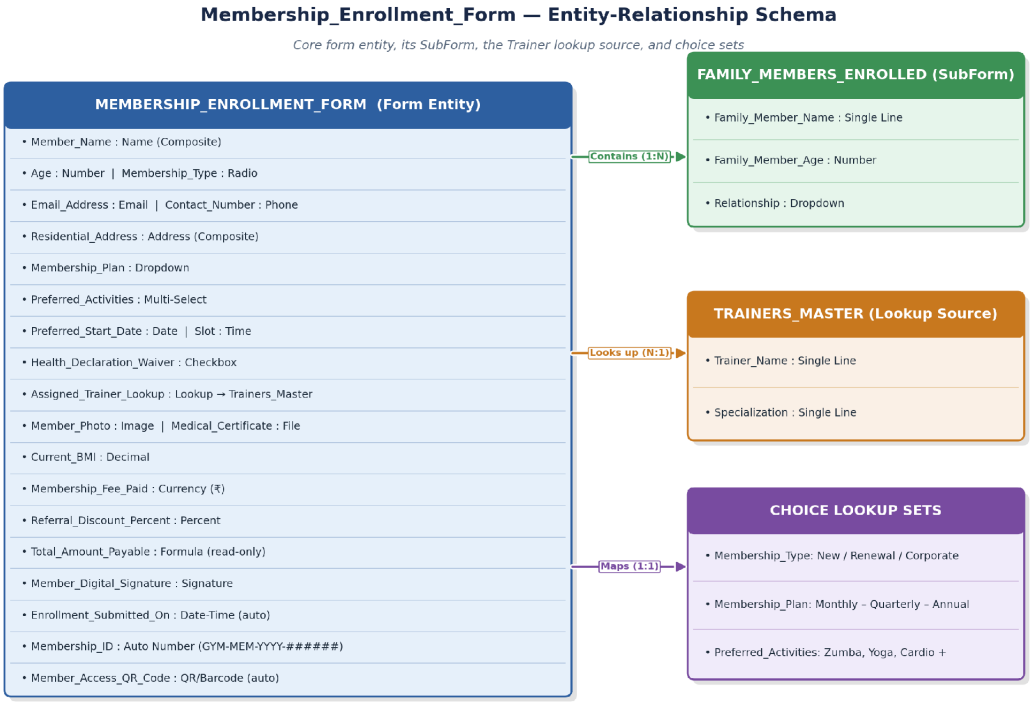


Figure 1: Entity-Relationship schema for Membership_Enrollment_Form

2.3 Basic Fields Mapping & Data Dictionary

Table 1 maps all thirteen Basic Fields from the Zoho Creator Form Builder palette to their role in the form, along with sample values drawn from the actual test submissions used while building this application.

Basic Field	Form Field Name	Field Link Name	Purpose	Sample Value
Name	Member Name	Member_Name	Captures full name using First/Last sub-fields (Mandatory).	Karthik Arjun
Number	Age	Age	Numeric age, validated to a 14–75 range (Mandatory).	24
Radio	Membership Type	Membership_Type	Single mandatory choice: New Member / Renewal / Corporate.	New Member
Email	Email Address	Email_Address	Validated email for confirmation and schedule updates (Mandatory).	karthich@gmail.com
Phone	Contact Number	Contact_Number	Validated mobile number, India (+91) format (Mandatory).	+91 98765 43210
Address	Residential Address	Residential_Address	Full address with city, state, and pincode.	14, Bharathi Nagar, Chennai, TN, 600028
Single Line	Emergency Contact Name	Emergency_Contact_Name	Short free-text entry for an emergency contact person.	Priya Raja (Spouse)
Multi Line	Fitness Goals / Medical Notes	Fitness_Goals_Medical_Notes	Free-text description of goals and relevant medical conditions.	Weight loss focus; mild knee discomfort.
Drop Down	Membership Plan	Membership_Plan	Single-select list of plans offered (Mandatory).	Quarterly Plan
Multi Select	Preferred Activities	Preferred_Activities	One or more fitness activities or classes to enroll in.	Weight Training, Zumba, Yoga
Date	Preferred Start Date	Preferred_Start_Date	Preferred membership start date; past dates disallowed.	01-Sep-2026
Time	Preferred Workout Slot	Preferred_Workout_Slot	Preferred daily time slot for workouts.	6:30 AM
Checkbox	Health Declaration / Waiver	Health_Declaration_Waiver	Declares fitness to exercise and accepts the liability waiver (Mandatory).	Checked (Agreed)

Table 1: Mapping and specification of the thirteen Basic Fields in Membership_Enrollment_Form

2.4 Advanced Fields Mapping & Data Dictionary

Table 2 extends the mapping to all seventeen Advanced Fields, which move the form beyond basic data capture into automation, computed values, file/media handling, and structured lookups.

Advanced Field	Form Field Name	Purpose	Sample Value / Behaviour
Decision Box	Requires Personal Trainer	Quick Yes/No on whether a personal trainer is wanted.	Yes / No toggle
Rich Text	Gym Rules Display	Read-only formatted gym rules shown inline on the form.	“Carry a towel at all times. Re-rack weights after use.”
URL	Fitness Social Profile URL	Link to the member's fitness-tracking or Instagram profile.	https://instagram.com/arjunfitness
Image	Member Photo	Passport-size photo for the gym access card.	member_photo.jpg (JPG/PNG, max 2 MB)
Percent	Referral Discount Percent	Discount applied when joining through a referral.	10% (range 0–20)
Currency	Membership Fee Paid	Membership fee amount, formatted in INR.	₹25,000.00
Decimal	Current BMI	Member's current Body Mass Index from the fitness assessment.	24.6

Date-Time	Enrollment Submitted On	Auto-captured exact date/time of submission (read-only).	16-Aug-2026 12:33:03
SubForm	Family Members Enrolled	Repeating rows of family member name, age, and relationship (0–5 rows).	Priya Kumar – 21 – Spouse
Lookup	Assigned Trainer Lookup	Type-ahead search against a master Trainers_Master table; auto-fills specialization.	Karthik M. – Strength & Conditioning
Signature	Member Digital Signature	Captures a drawn signature acknowledging the waiver (Mandatory).	Signed (drawn signature captured)
File Upload	Medical Certificate Upload	Scanned medical fitness certificate, if required.	medical_certificate.pdf (PDF/JPG/PNG, max 5 MB)
Audio	Health Concern Voice Note	Optional short voice note describing health concerns.	voice_note.mp3 (optional)
Video	Fitness Goal Video Intro	Short video stating fitness goals, for the trainer's reference.	goal_intro.mp4 (max 60 sec, 50 MB)
Auto Number	Membership ID	Unique, sequential membership number generated on submit.	GYM-MEM-2026-000001 (prefix + year + 6-digit sequence)
Formula	Total Amount Payable	Auto-computed final amount from Fee − (Fee × Discount), read-only.	₹22,500.00 (from ₹25,000 at 10% off)
QR/Barcode	Member Access QR Code	Scannable QR encoding the Membership ID, for gate access.	QR image encoding the Membership ID

Table 2: Mapping and specification of the seventeen Advanced Fields in Membership_Enrollment_Form

2.5 Form Layout & Configuration

The form was organized into eight logical, member-friendly sections, the first four built with Basic Fields and the remaining four extending the form with Advanced Fields:

Section A — Member Details: Member Name (Name), Age (Number), Membership Type (Radio).

Section B — Contact Details: Email Address (Email), Contact Number (Phone), Residential Address (Address), Emergency Contact Name (Single Line).

Section C — Membership & Fitness Preferences: Membership Plan (Drop Down), Preferred Activities (Multi Select), Fitness Goals / Medical Notes (Multi Line).

Section D — Scheduling & Health Declaration: Preferred Start Date (Date), Preferred Workout Slot (Time), Health Declaration / Waiver (Checkbox).

Section E — Fitness Assessment & Trainer Assignment: Assigned Trainer Lookup, Member Photo, Medical Certificate Upload, Current BMI.

Section F — Family & Media Details: Family Members Enrolled (SubForm), Fitness Social Profile URL, Fitness Goal Video Intro, Health Concern Voice Note.

Section G — Fee & Trainer Preference: Requires Personal Trainer (Decision Box), Membership Fee Paid (Currency), Referral Discount Percent (Percent), Total Amount Payable (Formula, read-only).

Section H — Declaration & System-Generated Values: Gym Rules Display (Rich Text), Member Digital Signature (Signature), Enrollment Submitted On (Date-Time), Membership ID (Auto Number), Member Access QR Code (QR/Barcode).

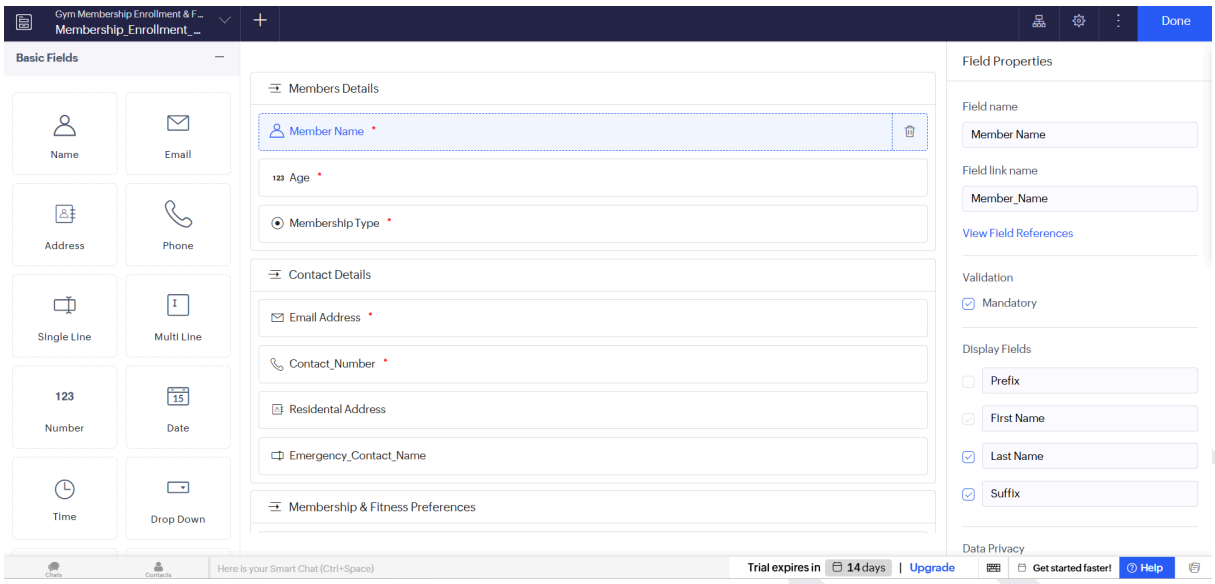


Figure 2: Form Builder workspace showing the thirteen Basic Fields across Sections A–D, with the Field Properties inspector open

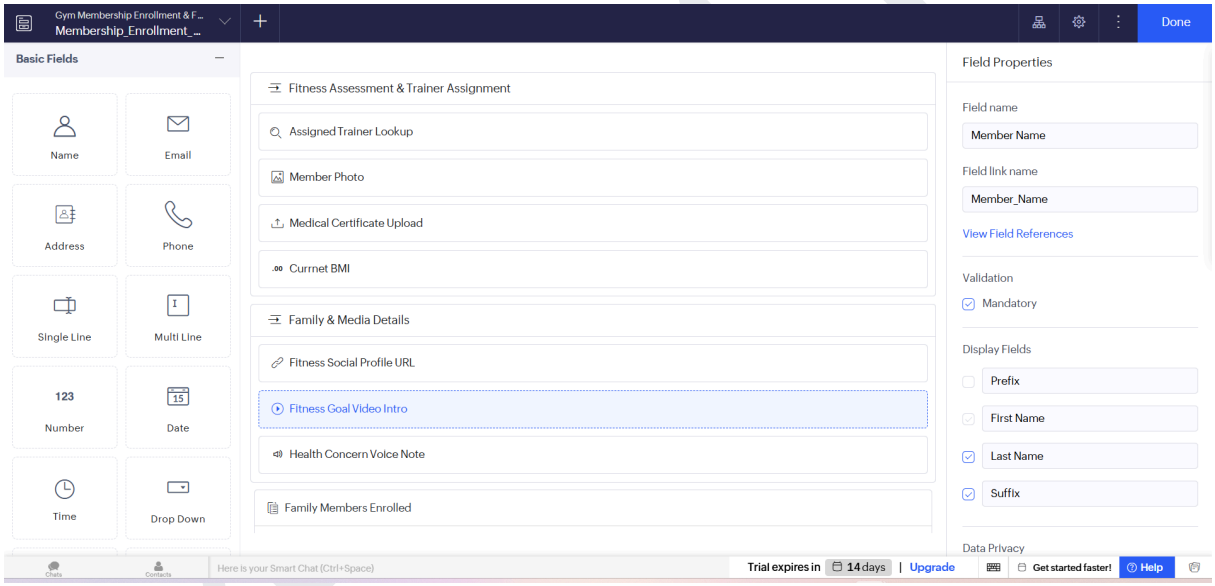


Figure 3: Form Builder canvas showing the seventeen Advanced Fields across Sections E–H

2.6 Workflow Scripting Configuration

A Deluge validation script was configured under Workflows > Form Workflows > Validations on form submission to enforce three business-logic checks before a record is accepted: the member's age must fall within the 14–75 range, the Preferred Start Date cannot be in the past, and the Health Declaration / Waiver checkbox must be ticked.

```
if(input.Age < 14 || input.Age > 75)
{
	alert "Age must be between 14 and 75.";
	cancel submit;
}
if(input.Preferred_Start_Date < zoho.currentdate)
{
	alert "Preferred Start Date cannot be in the past.";
	cancel submit;
}
if(input.Health_Declaration_Waiver == null || input.Health_Declaration_Waiver.isEmpty())
```

```
{
        alert "You must accept the health declaration and waiver.";
        cancel submit;
}
```

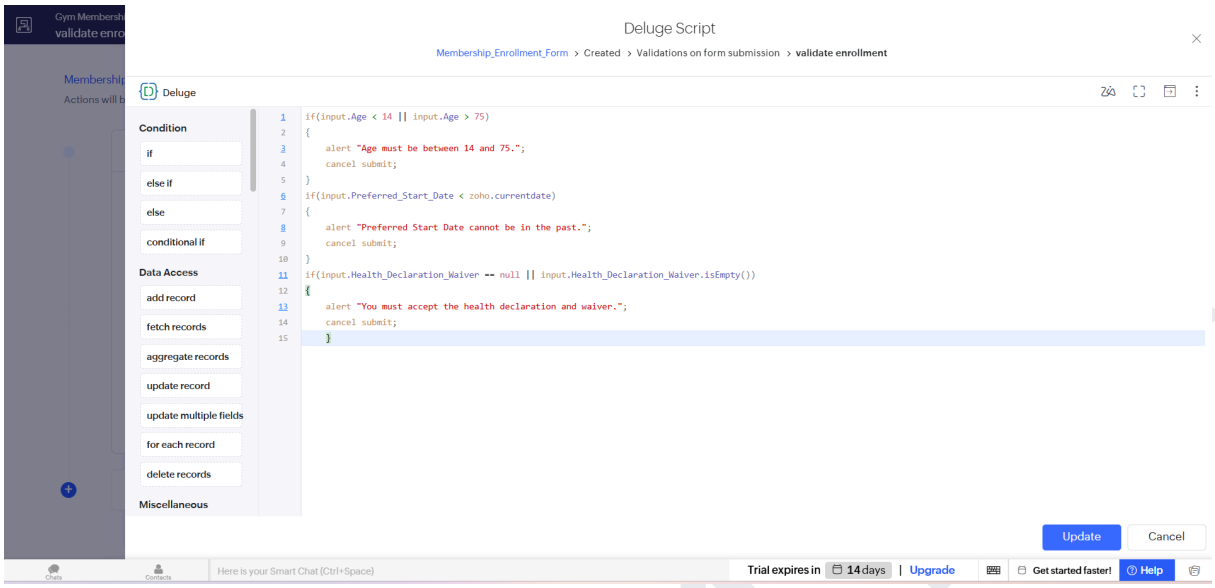


Figure 4: Deluge validation script canvas (Validate Enrollment — On Validate event)

3. Output

The following figures show the live preview of the form — including the SubForm and the computed Formula field — and the submitted member records in the report view.

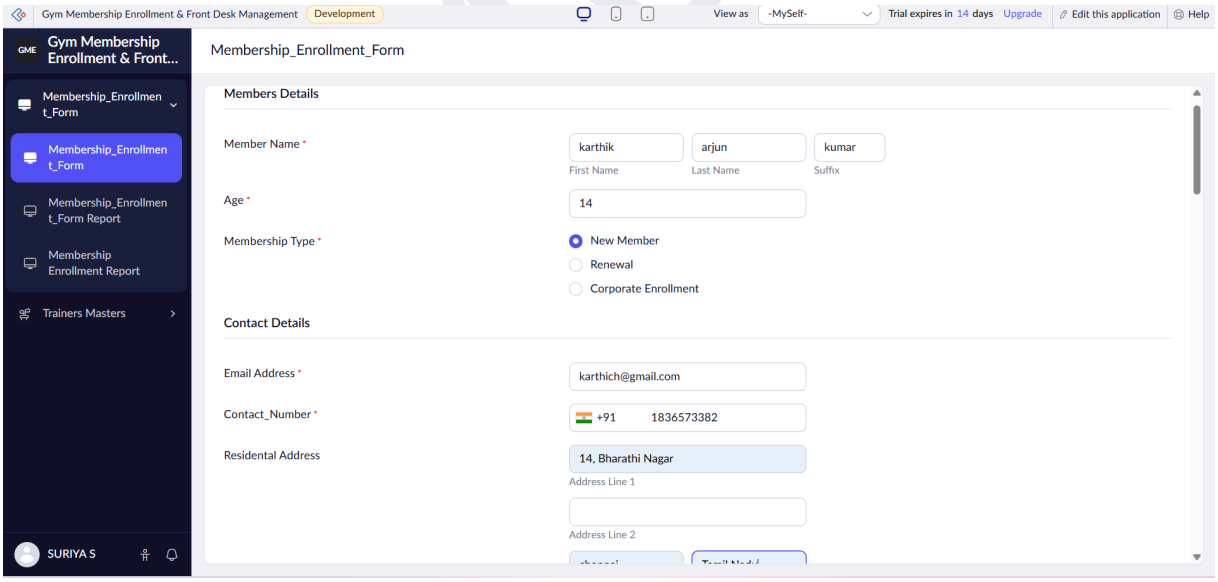


Figure 5: Live application runtime preview — basic-field sections filled with sample member data

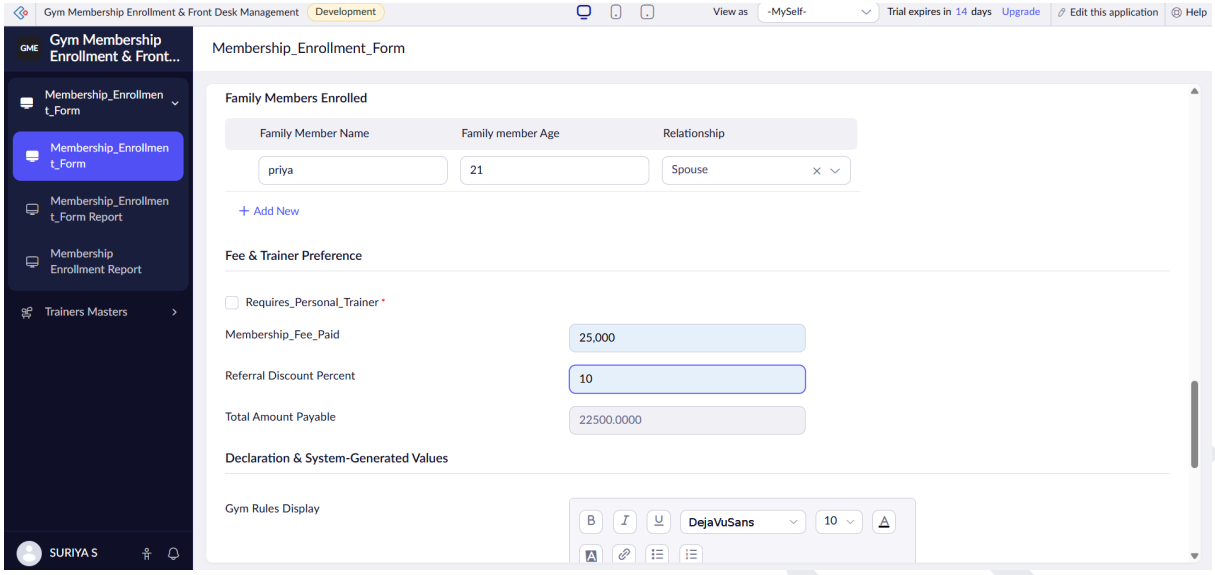


Figure 6: Live preview showing the Family Members Enrolled SubForm and the auto-computed Total Amount Payable

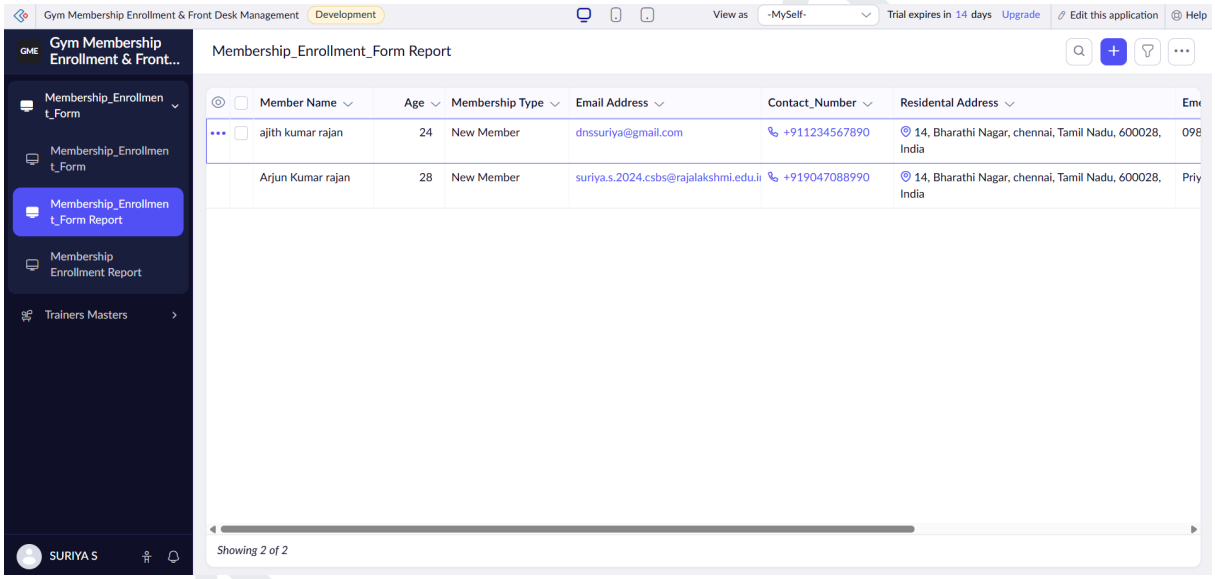


Figure 7: Membership_Enrollment_Form Report — list view of submitted member records

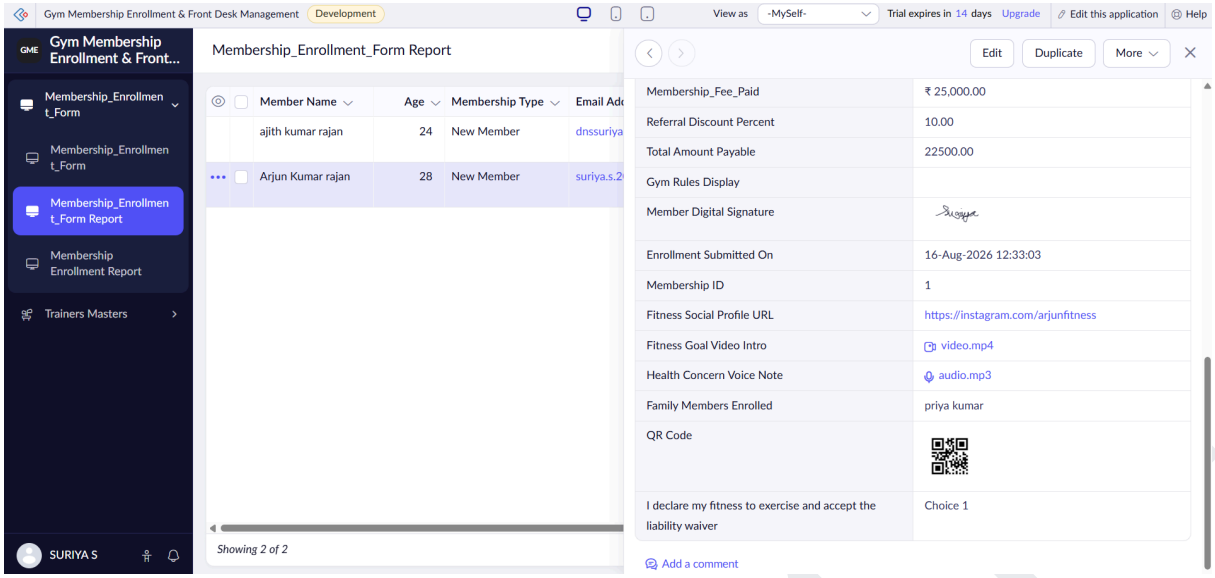


Figure 8: Submitted record detail showing the auto-generated Membership ID, Total Amount Payable, digital signature, and Member Access QR Code

4. Result

The "Gym Membership Enrollment Form" application was successfully built in Zoho Creator using all thirteen Basic Fields and all seventeen Advanced Fields, with field-level validations, an eight-section layout, a working SubForm for family enrollment, an auto-computed fee formula, a Trainer Lookup against a master table, and automated Membership ID / QR code generation — meeting all assignment requirements for both the basic and advanced field use cases.

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